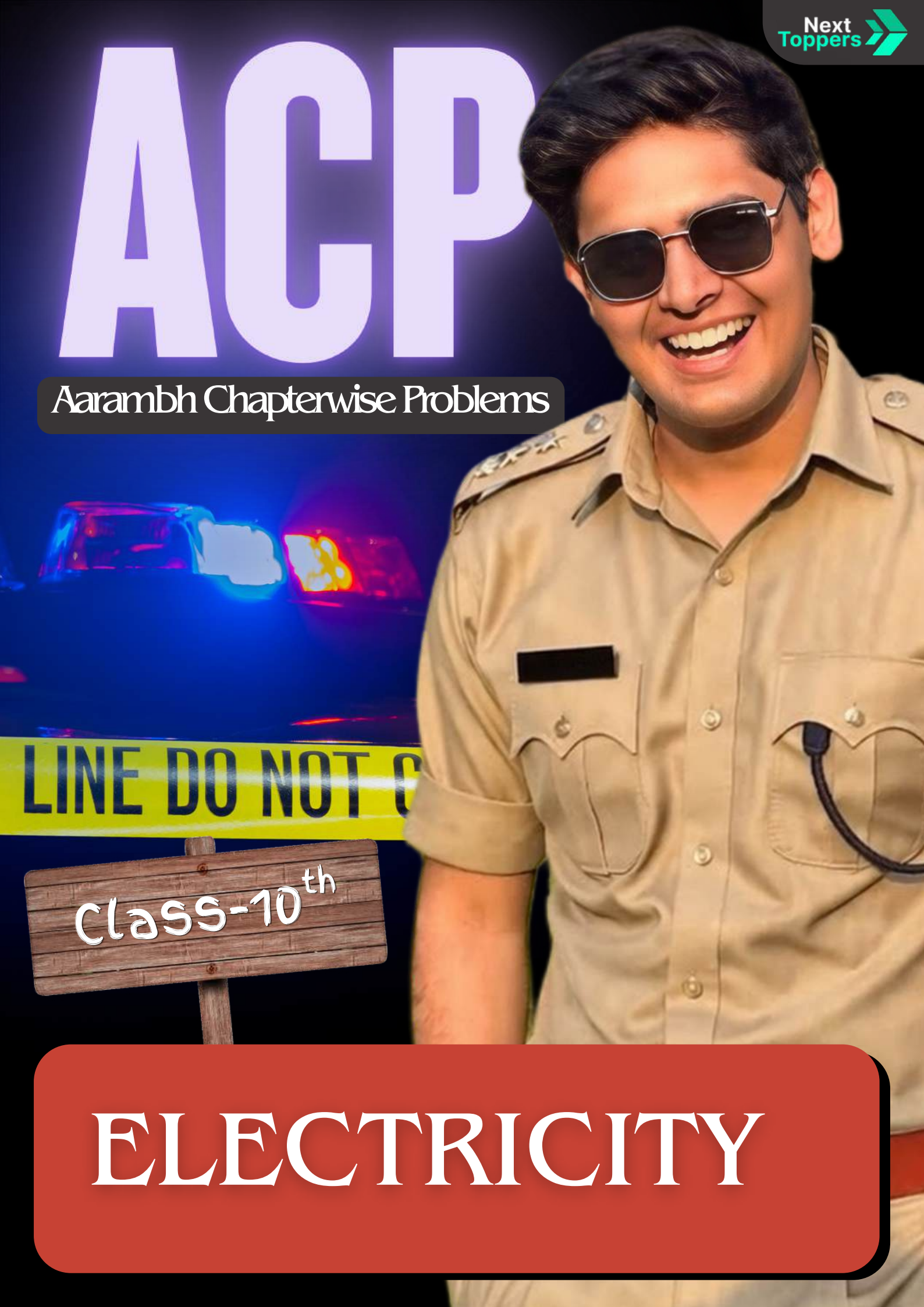


ACP

Aarambh Chapterwise Problems



LINE DO NOT CROSS

class-10th

ELECTRICITY

ELECTRICITY

PART 1 (Basic MCQs)

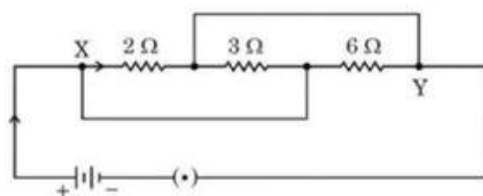
Read the question properly!

1. In case of four wires of the same material, the resistance will be minimum if the diameter and length of the wire respectively are:

- (a) $D/2$ and $L/4$
- (b) $D/4$ and $4L$
- (c) $2D$ and L
- (d) $4D$ and $2L$

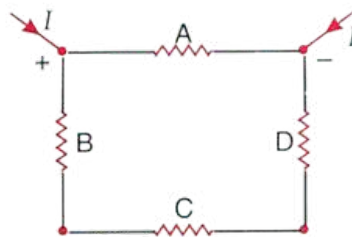
2. In the given circuit, the total resistance between X and Y is:

- (a) $12\ \Omega$
- (b) $4\ \Omega$
- (c) $6\ \Omega$
- (d) $1\ \Omega$



3. The diagram shows a network of four resistors A, B, C, and D connected to an electric source. Which of the following resistors are connected in series?

- (a) A, B and C
- (b) B, C and D
- (c) A, C and D
- (d) A, B and D



4. A cylindrical copper wire X of length L and radius r has resistance R and resistivity ρ . Another copper wire Y has length $2L$ and radius $2r$. Which of the following rows in the table shows the correct resistance and resistivity for copper wire Y?

- (a) Resistance = $R/2$, Resistivity = ρ
- (b) Resistance = R , Resistivity = ρ
- (c) Resistance = $R/2$, Resistivity = $\rho/2$
- (d) Resistance = R , Resistivity = $\rho/2$

5. The resistance of a resistor is reduced to half of its initial value while the potential difference across it and the time of current flow remain unchanged. The heat produced in the resistor will become:

- (a) Two times
- (b) Half
- (c) One-fourth
- (d) Four times

6. At the time of short circuit, the electric current in the circuit:

- (a) vary continuously.
- (b) does not change.
- (c) reduces substantially.
- (d) increases heavily.

**JOSH
METER**



7. Two bulbs of $100\ \Omega$ and $40\ \Omega$ are connected in series. The current through the $100\ \Omega$ bulb is 1 A . The current through the $40\ \Omega$ bulb will be:

- (a) 0.4 A
- (b) 0.6 A
- (c) 0.8 A
- (d) 1 A

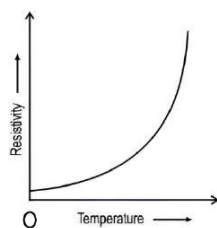
8. When a 4 V battery is connected across an unknown resistor there is current of 100 mA in the circuit. The value of the resistance of the resistor is:

- (a) $4\ \Omega$
- (b) $40\ \Omega$
- (c) $400\ \Omega$
- (d) $0.4\ \Omega$

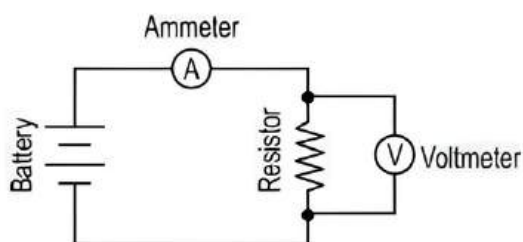
9. The equivalent resistance of a series combination of two resistances is $X\ \Omega$. If the resistances are $10\ \Omega$ and $40\ \Omega$ respectively, the value of X will be:

- (a) $10\ \Omega$
- (b) $20\ \Omega$
- (c) $50\ \Omega$
- (d) $40\ \Omega$

10. The graph below shows the variation of resistivity of copper with temperature.



Kishore constructs a simple circuit as shown. The resistor is made of copper.



He then heats the copper resistor. What will happen to the current flowing through the circuit? Why?

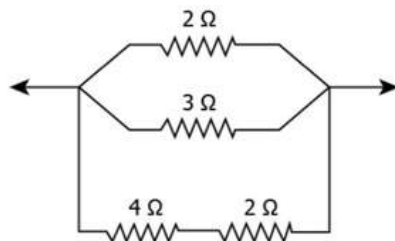
- (a) The current will increase because the resistance of copper increases with an increase in temperature.
- (b) The current will increase because the resistance of copper decreases with an increase in temperature.
- (c) The current will decrease because the resistance of copper increases with an increase in temperature.
- (d) The current will decrease because the resistance of copper decreases with an increase in temperature.

11. When too many electrical appliances are connected to a single socket in a circuit having a fixed value of current rating:

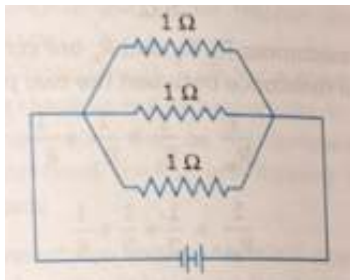
- (a) the total resistance of the circuit increases.
- (b) the current drawn from the source decreases.
- (c) the current increases beyond the rated value.
- (d) each appliance starts drawing more and more current from the source.

12. The image shows a combination of 4 resistors. What is the net resistance between the two points in the circuit?

- (a) $0.5\ \Omega$
- (b) $1.0\ \Omega$
- (c) $1.5\ \Omega$
- (d) $2.0\ \Omega$



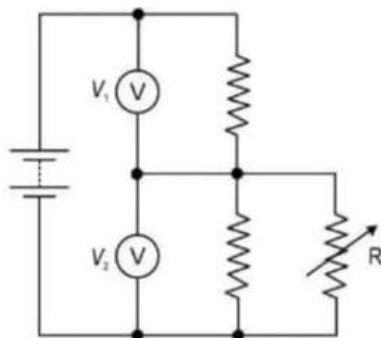
13. Three resistors are connected in a parallel combination in the circuit.



What is the equivalent resistance of the circuit?

- (a) Less than $1\ \Omega$
- (b) $1\ \Omega$
- (c) $2\ \Omega$
- (d) More than $2\ \Omega$

14. The variable resistor R is connected in a circuit, as shown in the figure. A variable resistor is one whose resistance can be changed.



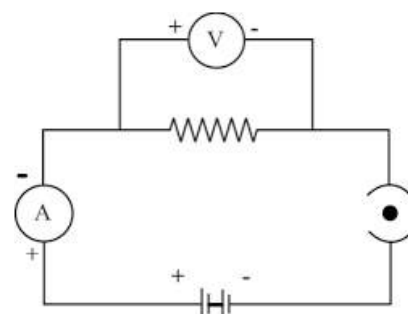
The resistance of R is increased. What will happen to the voltmeter readings V_1 and V_2 ?

- (a) V_1 increases, V_2 increases
- (b) V_1 increases, V_2 decreases
- (c) V_1 decreases, V_2 increases
- (d) V_1 decreases, V_2 decreases

15. The picture shows an electric circuit. Which of these is true about the circuit?

- (i) The circuit is open.
 - (ii) The circuit has two batteries.
 - (iii) The circuit has an ammeter and a voltmeter connected to each other.
- Options:

- (a) Only (i)
- (b) Only (ii)
- (c) Only (iii)
- (d) (i) and (ii)



ASSERTION AND REASON TYPE QUESTION (1 Marks)

*iyaha Marks katate hi*

In the following questions a statement of Assertion is followed by a statement of Reason.

Mark the correct choice as two statements are given one labeled Assertion (A) and the other labeled Reason (R).

Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:

- (a) Both A and R are true, and R is correct explanation of the assertion.
- (b) Both A and R are true, but R is not the correct explanation of the assertion.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

16. Assertion (A): Two wires of the same material have equal length, but one has twice the radius of the other. The thicker wire offers one-fourth the resistance of the thinner wire.

Reason (R): Resistance is inversely proportional to the cross-sectional area of the conductor.

17. Assertion (A): A 200 W, 220 V electric heater has lower resistance than a 100 W, 220 V heater.

Reason (R): At the same supply voltage, the resistance of an appliance decreases as its power rating increases.

18. Assertion (A): Replacing copper connecting wires with nichrome wires in a household circuit is not advisable.

Reason (R): Nichrome has much higher resistivity than copper.

19. Assertion (A): The equivalent resistance of three identical resistors connected in parallel is always less than the resistance of any one resistor.

Reason (R): Adding more parallel branches provides additional paths for current flow.

20. Assertion (A): If the diameter of a wire is reduced to half while keeping its length unchanged, its resistance becomes four times.

Reason (R): The area of cross-section is proportional to the square of the diameter.

21. Assertion (A): When resistors are connected in series, the potential difference across each resistor depends on its resistance.

Reason (R): In a series circuit, the same current flows through every resistor.

22. Assertion (A): Two resistors of $4\ \Omega$ and $12\ \Omega$ connected in parallel draw different currents from the same battery.

Reason (R): The current through a resistor connected across a fixed voltage is inversely proportional to its resistance.

23. Assertion (A): The resistance of a metallic conductor increases when it is heated.

Reason (R): Heating increases lattice vibrations, causing more collisions for moving electrons.

24. Assertion (A): A fuse wire is never connected in parallel with an electrical appliance.

Reason (R): A fuse must carry the entire circuit current to protect the appliance.

25. Assertion (A): Two identical bulbs connected in series glow dimmer than a single bulb connected to the same battery.

Reason (R): The current flowing through the series circuit is smaller due to higher equivalent resistance.

PART 2 (Subjective Questions)

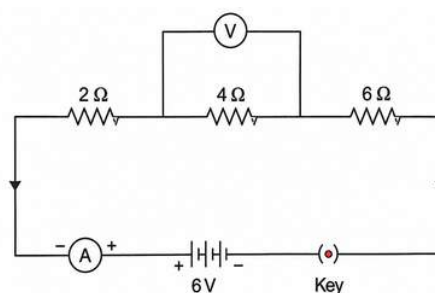
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SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

26. (A) State the relationship between the resistance R of a wire to its length l and cross-sectional area A . Use the mathematical symbols to arrive at the final formula.

(B) Using the formula, define the resistivity of a material.

27. In the given circuit, when the key is closed, determine the following:



(A) Total resistance of the circuit.

(B) Reading of the ammeter.

(C) Reading of the voltmeter.

28. Ananya responded to the question: Why are electrical appliances with metallic bodies connected to the mains through a three-pin plug, whereas an electric bulb can be connected with a two-pin plug?

She wrote: "Three-pin connections reduce heating of connecting wires."

(A) Is her answer correct or incorrect? Justify.

(B) What is the function of a fuse in a domestic electric circuit?

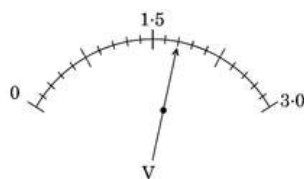
29. A student needs to make a 0.12Ω resistor. She has some copper wire of 0.80 mm diameter. Resistivity of copper is $1.8 \times 10^{-8} \Omega \text{ m}$.

(A) Determine the cross-sectional area of the wire.

(B) Calculate the length of wire required for the 0.12Ω resistor.

(C) Show that the length of the wire is approximately 3.33 m .

30. Consider the scale of a voltmeter shown in the figure and answer the following questions:



(a) What is the least count of the voltmeter?

(b) What is the value indicated by the pointer?

(c) If the voltmeter is connected across a 20Ω resistor, calculate the current flowing through the resistor.

(d) If this resistor is connected to a 20Ω resistor in series, how much energy will be consumed in one hour?

31. A current of 5 A is flowing through an electric heater for 30 minutes . The heater is connected to a 240 V supply.

(a) Calculate the electrical energy consumed by the heater in 30 minutes .

(b) Express the energy consumed in kilowatt-hour (kWh).

(c) If electricity costs ₹8 per unit, calculate the cost of operating the heater.

32. Two wires, P and Q, are made of the same material and have the same cross-sectional area. The length of wire P is 50 cm, while the length of wire Q is 150 cm.

- Which wire has greater resistance?
- Calculate the ratio of their resistances, $R_P : R_Q$
- Explain how the resistance of a conductor depends on its length.

33. A mobile charger consumes an average power of 30 W and is used continuously for 30 days. Calculate the electrical energy consumed in one month.

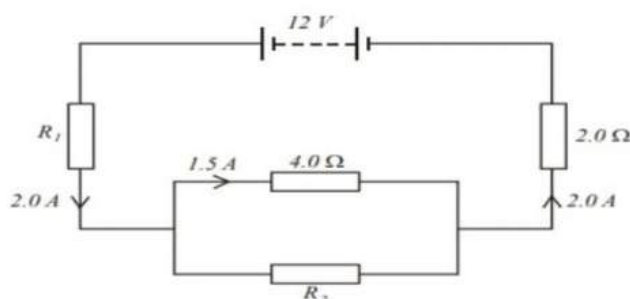
34. A conductor has a resistance of $10\ \Omega$.

- On what factors does the resistance of a conductor depend?
- George says that the resistance of a conductor depends on the material of the conductor. Explain.
- If two similar wires of the same length and diameter are made of copper and nichrome, the nichrome wire offers much greater resistance. Explain why.

LONG ANSWER TYPE QUESTIONS (5 Marks)

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35.



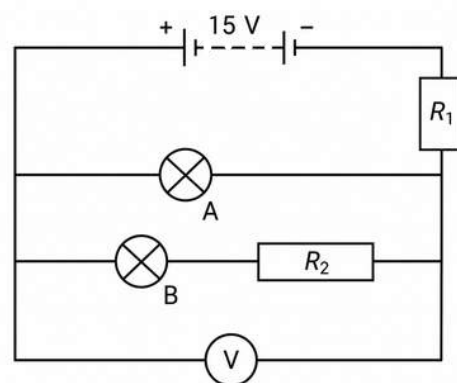
The above circuit is part of an electrical appliance. Use the information given in the circuit to calculate the following:

- Potential difference across R_2 .
- Value of resistance R_2 .
- Value of resistance R_1 .

36. As shown in the given figure, A and B are two lamps. Lamp A is rated at 12 V, 24 W. Lamp B is rated at 6.0 V. When lamp B operates at its rated voltage, the current flowing through it is 3.0 A. The values of R_1 and R_2 are chosen so that both lamps operate at their rated voltages.

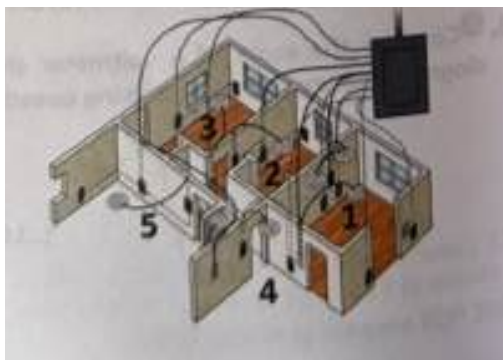
Based on the information given, answer the following:

- Calculate the current in lamp A.
- State the reading of the voltmeter and give a reason for your answer.
- Calculate the resistance of R_2 .
- Find the value of resistance R_1 .



37. The given diagram is a schematic diagram of a house showing four appliances connected in different rooms. For this house:

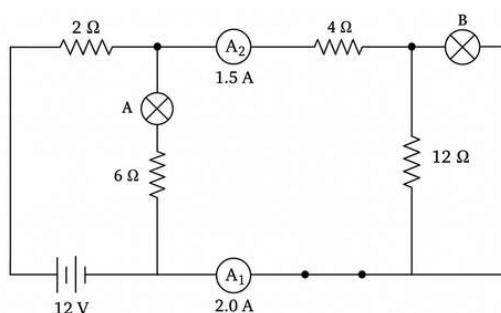
- Main supply = 220 V
- Net current from the mains = 5.2 A



Answer the following:

- (A) Calculate the power consumed by the house.
- (B) Calculate the electrical energy consumed in 30 days if all appliances are used as shown.
- (C) State why household appliances are connected in parallel.

38.



A student performed an experiment using the circuit shown. The circuit contains two bulbs, three resistors and ammeters.

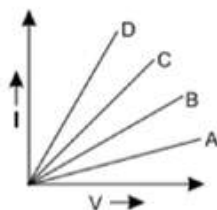
Answer the following:

- (A) What is the resistance of bulb B?
- (B) Calculate the current flowing through bulb B.
- (C) Calculate the equivalent resistance of the circuit.

PART 3 (Advanced Questions)

Case Study/Source Based Question (5 Marks)

39. Ohm's law gives a relationship between current and potential difference. According to this law, at constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends. The ratio of potential difference applied between the ends of a conductor and the current flowing through it is a constant quantity called resistance. The following graph is obtained by a researcher while doing an experiment to study Ohm's law. The I–V graph for four conductors A, B, C and D having resistances R_A , R_B , R_C and R_D respectively are shown in the graph.



i. If all the conductors are of same length and same material, which is the thickest?

- (a) C
- (b) D
- (c) A
- (d) B

ii. If all the conductors are of same thickness and of same material, which is the longest?

- (a) B
- (b) C
- (c) A
- (d) D

iii. Which one of the following relations is true for these conductors?

- (a) $R_A > R_B > R_C > R_D$
- (b) $R_A = R_B < R_C < R_D$
- (c) $R_A < R_B < R_C < R_D$
- (d) $R_A = R_B = R_C = R_D$

iv. If conductors A and B are connected in series and I–V graph is plotted for the combination, its slope would be:

- (a) more than that of A
- (b) between A and B
- (c) more than that of D
- (d) less than that of A

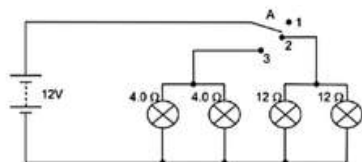
Or

iv. If conductors C and D are connected in parallel and I–V graph is plotted for the combination, its slope would be:

- (a) between C and D
- (b) lesser than that of A
- (c) more than that of D
- (d) between B and C

40. Read the source below and answer the questions that follow:

Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition.



i. State what happens when switch A is connected to (1 mark)

- (a) Position 2
- (b) Position 3

ii. Find the potential difference across each lamp when lit. (1 mark)

iii. Calculate the current:

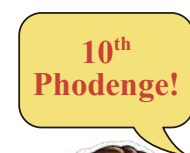
- (a) In each $12\ \Omega$ lamp when lit.
- (b) In each $4\ \Omega$ lamp when lit.

iv. Show, with calculations, which type of lamp, $4.0\ \Omega$ or $12\ \Omega$ has the higher power.

41. When an electric current passes through a conductor (like a high resistance wire) the conductor becomes hot after some time and produces heat. This is called heating effect of electric current or joule heating or ohmic heating or resistive heating. The heating effect of electric current is widely used in our day-to-day life. The electric laundry iron, electric toaster, electric bulb, fuse and electric heater are some of the familiar devices based on Joule's heating.

i. Write the mathematical expression for Joule's law of heating. (1 mark)

ii. The following table gives the resistivity of three samples:



Sample	A	B	C
Resistivity	$1.6 \times 10^{-5} \Omega \text{ m}$	$5.2 \times 10^{-5} \Omega \text{ m}$	$100 \times 10^{-4} \Omega \text{ m}$

Which of them is suitable for heating elements of electrical appliances and why?

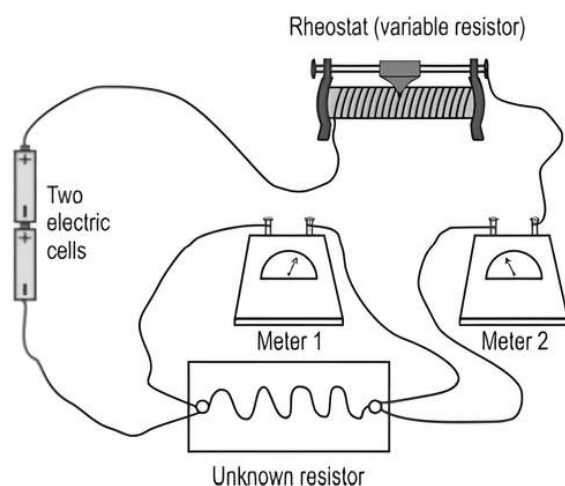
iii. Why does the cord of an electric heater not glow while the heating element does? (1 mark)

iv. State a difference between the wire used in the element of an electric heater and in a fuse wire. (1 mark)

Or

iv. Tungsten is used almost exclusively for filaments of electric bulb. List two reasons.

42. The diagram below shows how amita had connected a circuit to verify ohm's law.



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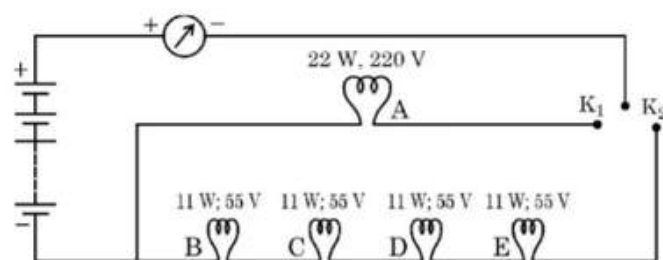
(a) Identify which of the device in the circuit is an ammeter. Justify your answer.

(b) Draw a circuit diagram with appropriate symbols for the circuit shown in the diagram above.

(c) Amita forgot to put a switch in the circuit. During the experiment, the wire labelled 'Unknown resistor' became hot. The resistivity of the material of the wire increases with temperature. Draw two potential difference (V) vs current (I) graphs (in the same diagram):

- (i) as expected by Amita,
- (ii) as based on the actual observation she would make.

43. In a domestic circuit, five LED bulbs are arranged as shown. The source voltage is 220 V and the power rating of each bulb is marked in the circuit diagram. Based on the following circuit diagram, answer the following questions:



(a) State what happens when

(i) key K_1 is closed.

(ii) key K_2 is closed.

(b) Find the current drawn by the bulb B when it glows.

(c) Calculate

(i) the resistance of bulb B, and

(ii) the total resistance of the combination of four bulbs B, C, D, and E.

OR

(c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 are both closed and the bulb C suddenly gets fused? Give a reason to justify your answer.

- Resistance, Resistivity & Factors Affecting Resistance – Q1, Q4, Q10, Q16, Q18, Q20, Q23, Q26, Q29, Q32, Q34
- Ohm's Law ($V = IR$) & V–I Graph – Q8, Q10, Q27, Q30, Q35, Q36, Q38, Q39, Q42
- Series & Parallel Combination of Resistors – Q2, Q3, Q7, Q9, Q12, Q13, Q19, Q21, Q22, Q27, Q35, Q36, Q38
- Heating Effect of Electric Current (Joule's Law) – Q5, Q17, Q31, Q41
- Electric Power & Electrical Energy – Q17, Q31, Q33, Q37, Q40
- Domestic Electric Circuits (House Wiring, Fuse, Earthing, Overloading & Short Circuit) – Q6, Q11, Q18, Q24, Q28, Q37, Q40, Q43
- Circuit Diagrams, Ammeter & Voltmeter – Q14, Q15, Q27, Q30, Q35, Q36, Q38, Q42, Q43
- Assertion & Reason – Q16–Q25
- Resistance & Resistivity Numericals – Q27, Q29, Q30, Q35, Q36, Q38
- Household Electricity & Safety Measures – Q6, Q11, Q24, Q28, Q37, Q40, Q43
- Long Answer Questions (5 Marks) – Q35, Q36, Q37, Q38
- Case Study – Ohm's Law & I–V Graph – Q39
- Case Study – Car Headlights, Parallel Circuit, Current & Power – Q40
- Case Study – Heating Effect of Electric Current (Joule's Law) – Q41
- Ohm's Law Experiment (Circuit Diagram & V–I Graph) – Q42
- Domestic LED Bulb Circuit – Q43



SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

- | | | |
|--------|--------|--------|
| 1.(d) | 11.(c) | 21.(a) |
| 2.(d) | 12.(b) | 22.(a) |
| 3.(b) | 13.(a) | 23.(a) |
| 4.(a) | 14.(c) | 24.(a) |
| 5.(a) | 15.(a) | 25.(a) |
| 6.(d) | 16.(a) | |
| 7.(d) | 17.(a) | |
| 8.(b) | 18.(a) | |
| 9.(c) | 19.(a) | |
| 10.(c) | 20.(a) | |

SHORT ANSWER TYPE QUESTIONS:

26. The resistance of a conductor depends upon its length and cross-sectional area.

(i) Resistance is directly proportional to the length of the conductor. $R \propto l$ This means that if the length of the conductor increases, its resistance also increases because electrons have to travel through a longer path.

(ii) Resistance is inversely proportional to the cross-sectional area of the conductor. $R \propto 1/A$ This means that if the cross-sectional area increases, the resistance decreases because more free electrons can flow through the conductor.

Combining the above two relations, $R \propto l/A$ Introducing the proportionality constant called resistivity (ρ), $R = \rho l/A$ This is the required relation between resistance, resistivity, length and cross-sectional area.

(B) Resistivity is the resistance of a conductor having unit length and unit cross-sectional area. It is a characteristic property of the material and depends on the nature of the material and temperature.

SI Unit: Ohm metre (Ω m).

27.(A) Total Resistance = $2\ \Omega + 4\ \Omega + 6\ \Omega = 12\ \Omega$

(B) Using Ohm's Law,

$$I = V/R$$

$$= 6/12$$

$$= 0.5\text{ A}$$

Therefore, the ammeter reading is 0.5 A.

(C) Voltage across the $4\ \Omega$ resistor,

$$V = IR$$

$$= 0.5 \times 4$$

$$= 2\text{ V}$$

Therefore, the voltmeter reading is 2 V.

28. (A) No. The answer is incorrect.

A three-pin plug is used because the earth wire provides protection against electric shock. It connects the metallic body of the appliance to the earth. If any leakage current flows, it passes safely to the ground, preventing electric shock to the user.

(B) A fuse is a safety device used in an electric circuit. It protects electrical appliances from overloading and short circuit. When excessive current flows through the circuit, the fuse wire melts due to the heating effect of electric current and breaks the circuit, thereby protecting the appliance.

29.(A) Radius = $0.40\text{ mm} = 4 \times 10^{-4}\text{ m}$

Cross-sectional area, $A = \pi r^2$

$$= 3.14 \times (4 \times 10^{-4})^2$$

$$= 5.03 \times 10^{-7}\text{ m}^2$$

Therefore, the cross-sectional area of the wire is $5.03 \times 10^{-7}\text{ m}^2$.

(B) Using,

$$R = \rho l/A$$

$$l = RA/\rho$$

$$= (0.12 \times 5.03 \times 10^{-7})/(1.8 \times 10^{-8})$$

$$\approx 3.33\text{ m}$$

Therefore, the required length of the wire is approximately 3.33 m.

(C)

Hence, the length of the wire is approximately 3.33 m.

30.(A) Least count of the voltmeter = $3.0/20 = 0.15\text{ V}$

(B) Voltmeter reading = $1.5 + (2 \times 0.15) = 1.8\text{ V}$

(C) Using Ohm's Law,

$$I = V/R$$

$$= 1.8/20$$

$$= 0.09\text{ A}$$

Therefore, the current flowing through the resistor is 0.09 A.

(D) Two $20\ \Omega$ resistors connected in series:

$$R_{\text{total}} = 20 + 20 = 40\ \Omega$$

$$E = V^2/R$$

$$= \{(1.8)^2/40\} \times 3600$$

$$= 291.6\text{ J}$$

31.(A) Power, $P = VI$

$$= 240 \times 5$$

$$= 1200 \text{ W}$$

Energy, $E = Pt$

$$= 1200 \times 1800$$

$$= 2.16 \times 10^6 \text{ J}$$

Therefore, the electrical energy consumed is $2.16 \times 10^6 \text{ J}$.

(B) Power = 1.2 kW

Time = 0.5 h

Energy = Power $\times$ Time

$$= 1.2 \times 0.5$$

$$= 0.6 \text{ kWh}$$

Therefore, the electrical energy consumed is 0.6 kWh (0.6 unit).

(C) Cost = Energy $\times$ Cost per unit

$$= 0.6 \times 8$$

$$= ₹4.80$$

Therefore, the cost of operating the heater is ₹4.80.

32.(a) Wire O has greater resistance.

(b) Since

$$R \propto l$$

$$\frac{R_P}{R_Q} = \frac{l_P}{l_Q} = \frac{50}{150} = \frac{1}{3}$$

Therefore,

$$R_P : R_Q = 1 : 3$$

c) The resistance of a conductor is directly proportional to its length. A longer conductor offers greater resistance because electrons travel through a longer path and undergo more collisions.

33. Power = 30 W = 0.03 kW

Time = 30 days = 720 hours

Energy consumed = Power $\times$ Time

$$= 0.03 \times 720$$

$$= 21.6 \text{ kWh}$$

Therefore, the average electrical energy consumed in one month is 21.6 kWh or 21.6 units.

34.(A) The resistance of a conductor depends upon:

- Length of the conductor
- Cross-sectional area of the conductor
- Nature of the material (resistivity)
- Temperature of the conductor

(B)

Different materials have different values of resistivity. Therefore, conductors made of different materials have different resistances even if they have the same length and cross-sectional area.

(C)

Nichrome has much higher resistivity than copper. Since $R = \rho l/A$ the nichrome wire has greater resistance for the same length and cross-sectional area.

LONG ANSWER TYPE QUESTIONS:

35. Given:

- Battery voltage = 12 V
- Total current = 2.0 A
- Current through 4 Ω resistor = 1.5 A

(A) The 4 Ω resistor and R_2 are connected in parallel, so they have the same potential difference.

Using Ohm's Law:

$$V = IR$$

$$V = 1.5 \times 4 = 6 \text{ V}$$

Answer: The potential difference across R_2 is 6 V.

(B) Current through R_2 = Total current – Current through 4 Ω resistor

$$= 2.0 - 1.5$$

$$= 0.5 \text{ A}$$

Using Ohm's Law: $R = V \div I$

$$R = 6 \div 0.5 = 12 \Omega$$

Answer: Resistance of $R_2 = 12 \Omega$.

(C) $V = IR = 2 \times 2 = 4 \text{ V}$

Using the total supply voltage:

$$VR_1 = 12 - 6 - 4 = 2 \text{ V}$$

$$R_1 = V/I = 2/2 = 1\Omega$$

$$R_1 = 1\Omega$$

36. (A) For lamp A: $P = VI$

$$I = P/V = 24/12 = 2 \text{ A}$$

Therefore, the current in lamp A is 2 A.

(B) Voltmeter reading = 12 V

Reason: Lamp A and the branch containing lamp B are connected in parallel, so the potential difference across both branches is the same.

(C) Potential difference across R_2 :

$$V = 12 - 6 = 6 \text{ V}$$

Current through $R_2 = 3 \text{ A}$

$$R_2 = V/I = 6/3 = 2 \text{ ohm}$$

Therefore, $R_2 = 2 \text{ ohm}$.

(D) Current through R_1 :

$$I = 2 + 3 = 5 \text{ A}$$

Potential difference across R_1 :

$$V = 15 - 12 = 3 \text{ V}$$

$$R_1 = V/I = 3/5 = 0.6 \text{ ohm}$$

37. Given:

- Supply voltage = 220 V
- Current = 5.2 A

(A) Using the formula:

$$P = VI$$

$$P = 220 \times 5.2$$

$$P = 1144 \text{ W} = 1.144 \text{ kW}$$

(B) Time = 30×24

$$= 720 \text{ hours}$$

$$\text{Energy} = \text{Power} \times \text{Time}$$

$$= 1.144 \times 720 = 823.68 \text{ kWh}$$

- (C) 1. Every appliance receives the full supply voltage (220 V).
 2. Each appliance can be switched ON or OFF independently.
 3. If one appliance fails, the other appliances continue to work normally.

38 Given: Total current = 2 A, current through $4 \Omega = 1.5 \text{ A}$.

Voltage across 2Ω : $V = IR = 2 \times 2 = 4 \text{ V}$

Remaining voltage: $12 - 4 = 8 \text{ V}$

Voltage across 4Ω : $V = 1.5 \times 4 = 6 \text{ V}$

So, voltage across bulb B and 12Ω resistor: $8 - 6 = 2 \text{ V}$

(A) Current through 12Ω : $I = 2/12 = 1/6 \text{ A}$

Current through bulb B: $I_B = 1.5 - 1/6 = 4/3 \text{ A} = 1.33 \text{ A}$

Resistance of bulb B: $R_B = 2 \div (4/3) = 1.5 \Omega$

(B) Current through bulb B = 1.33 A

(C) Equivalent resistance:

$$R_{eq} = V/I = 12/2 = 6 \Omega$$

CASE STUDY/SOURCE BASED QUESTION:

39 i. (b) D

ii. (c) A

iii. (a) $R_A > R_B > R_C > R_D$

iv. (d) Less than that of A

OR

iv. (c) More than that of D

40 ① (a) When switch A is connected to position 2, the two 12 ohm lamps glow.

(b) When switch A is connected to position 3, the two 4 ohm lamps glow.

(ii) Since the lamps are connected in parallel across a 12 V battery:

Potential difference across each lamp = 12 V

(iii) (a) For each 12 ohm lamp:

$$I = V/R$$

$$I = 12/12 = 1 \text{ A}$$

(b) For each 4 ohm lamp:

$$I = V/R$$

$$I = 12/4 = 3 \text{ A}$$

(iv) For a 4 ohm lamp:

$$P = V^2/R$$

$$P = 12^2/4 = 36 \text{ W}$$

For a 12 ohm lamp:

$$P = V^2/R$$

$$P = 12^2/12 = 12 \text{ W}$$

Therefore, the 4 ohm lamp has higher power. It acts as the headlight, while the 12 ohm lamp acts as the sidelight.

41 i. $H = I^2 R t$

ii. Sample C is suitable because it has the highest resistivity and produces more heat.

iii. The heating element has high resistance, so it becomes red hot and glows. The connecting cord has very low resistance, so it does not glow.

iv. Heating element wire has a high melting point and high resistance, whereas a fuse wire has a low melting point and melts when excess current flows.

OR

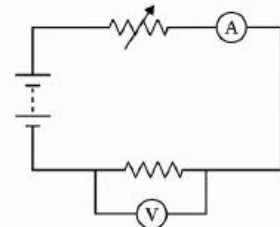
Reasons why tungsten is used as the filament of an electric bulb:

1. It has a very high melting point.
2. It has high resistivity and can glow white hot without melting.

42. (a) Meter 2, Because it is connected in series with the unknown resistor through which the current needs to be measured.

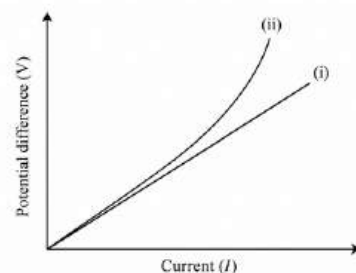
(b) Correct connection for the cell, the unknown resistor and the rheostat in the circuit diagram.

- Correct connections for the two meters in the diagram.
- Use of correct symbols for all components.



(c) (i) A straight-line V–I graph passing through the origin (as expected by Amita).

(ii) A curved V–I graph with an increasing slope (actual observation, since the resistance increases as the wire gets hotter).



43 (a) State what happens when:

(i) Key K_1 is closed. Only bulb A glows because its branch is completed.

(ii) Key K_2 is closed. Only bulbs B, C, D and E glow because their series branch is completed.

(b) Find the current drawn by bulb B when it glows.

Given:

$$\text{Power (P)} = 11 \text{ W}$$

$$\text{Voltage (V)} = 55 \text{ V}$$

Formula:

$$P = VI$$

$$I = P/V$$

$$= 11/55$$

$$= 0.2 \text{ A}$$

Answer: 0.2 A

(c) (i) Resistance of bulb B

$$\text{Formula: } R = V^2/P$$

$$R = (55 \times 55)/11$$

$$= 3025/11 = 275 \Omega$$

(ii) Total resistance of bulbs B, C, D and E

Since the four bulbs are connected in series,

$$R_{\text{total}} = 275 + 275 + 275 + 275 = 1100 \Omega$$

OR

(c) • Bulb A continues to glow normally.

• Bulbs B, C, D and E do not glow.

Reason: Bulbs B, C, D and E are connected in series. When bulb C gets fused, the circuit is broken, so no current flows through that branch. Bulb A is connected in a separate parallel branch, so it remains unaffected.